

Park City Experimental Mathematics Lab

On the Spectral Properties of $K_{m_1, \dots, m_r}$ and the Permutohedron

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Outline

Motivation The Kirchhoff matrix tree theorem says the number of spanning trees in a graph G is determined by the eigenvalues of G 's Laplacian matrix.

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Goal 1. Understand the number of spanning trees in $T_r(n)$ and $K_{m_1, \dots, m_r}$.

Goal 2. Understand the spectrum of the 1-skeleton of the permutohedron.

Progress

What's worked Using eigenvalues to enumerate spanning trees of

- $K_{n,n}$
- $T_r(n)$
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What hasn't Eigenvalues of the permutohedron have been harder to describe. However, we have been able to say a little about:

- the Laplacian matrix's recursive structure
- the limiting distribution of eigenvalues as n approaches infinity

Spanning Trees of $T_r(n)$

Proposition

The number of spanning trees of $T_r(mr)$ is $(m(r-1))^{mr-r}(mr)^{r-2}$.

Proof.

Define $G = T_r(n)$ and $\tilde{L}_G$ to be the Laplacian of G with the last row and column removed. Then,

$$\tilde{L}_G = m(r-1)I_{n-1} - \tilde{A}_G$$

where $\tilde{A}_G$ is the adjacency matrix of G with the last row and column removed. By Kirchhoff's matrix tree theorem:

$$\begin{aligned} \# \text{spanning trees } G &= \det \tilde{L}_G \\ &= \det \left(m(r-1)I_{n-1} - \tilde{A}_G \right) \end{aligned}$$

Proofs

Where the last expression is characteristic polynomial of $\tilde{A}_G$ at $m(r-1)$.
Find the characteristic polynomial by finding eigenvalues of $\tilde{A}_G$.

$$\tilde{A}_G = \left[\begin{array}{ccccc} \underbrace{0}_{m \times m} & \underbrace{1}_{m \times m} & \underbrace{1}_{m \times m} & \cdots & \underbrace{1}_{m \times (m-1)} \\ \underbrace{1}_{m \times m} & \underbrace{0}_{m \times m} & \underbrace{1}_{m \times m} & \cdots & \underbrace{1}_{m \times (m-1)} \\ \vdots & \vdots & \vdots & \ddots & \\ \underbrace{1}_{(m-1) \times m} & \underbrace{1}_{(m-1) \times m} & \underbrace{1}_{(m-1) \times m} & \cdots & \underbrace{0}_{(m-1) \times (m-1)} \end{array} \right] \left. \vphantom{\begin{array}{ccccc} \underbrace{0}_{m \times m} \\ \underbrace{1}_{m \times m} \\ \vdots \\ \underbrace{1}_{(m-1) \times m} \end{array}} \right\} r \text{ groups}$$

By row reducing rows in each of the r groups we get $\text{rank } \tilde{A}_G \leq r$.

Proofs

Guess a plausible general form for the eigenvector

$$\mathbf{v} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_r \end{bmatrix} \begin{matrix} m \\ m \\ \vdots \\ m-1 \end{matrix}$$

and deduce eigenvalues by solving

$$\tilde{A}_G \mathbf{v} = \lambda \mathbf{v}$$

Proofs

Case 1: $\exists a_q \neq a_t : q, t \leq r - 1$. We then have

$$m \sum_{\substack{i=1 \\ i \neq t}}^r a_i - a_r = \lambda a_t$$

$$m \sum_{\substack{i=1 \\ i \neq q}}^r a_i - a_r = \lambda a_q$$

which gives $\lambda = -m$.

Proofs

We explicitly construct $r - 2$ linearly independent eigenvectors corresponding to $\lambda = -m$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} m \\ m \\ m \\ \vdots \\ m-1 \end{matrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} m \\ m \\ m \\ \vdots \\ m-1 \end{matrix}, \quad \dots, \quad \mathbf{v}_{r-2} = \begin{bmatrix} 1 \\ \vdots \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{matrix} m \\ \vdots \\ m \\ m \\ m-1 \end{matrix}$$

which gives

$$\dim [E_{-m}(\tilde{A}_G)] \geq \dim [\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{r-2}\}] = r - 2.$$

Proofs

Case 2: $a_1 = \dots = a_{r-1}$. In this case, we only have 2 different equations:

$$m(r-2)a_1 + (m-1)a_r = \lambda a_1$$

$$m(r-1)a_1 = \lambda a_r$$

$\lambda = 0$ gives a trivial zero vector. Then, if $\lambda \neq 0$, we get

$$\lambda_1 = \frac{m(r-2) - \sqrt{(m(r-2))^2 + 4(r-1)m(m-1)}}{2} \neq 0$$

$$\lambda_2 = \frac{m(r-2) + \sqrt{(m(r-2))^2 + 4(r-1)m(m-1)}}{2} \neq 0$$

Proof

Both λ_1 and λ_2 have one-dimensional eigenspaces.

$\tilde{A}_G$ is symmetric, so geometric multiplicity is equal to algebraic multiplicity. Rank is the sum of algebraic multiplicities of non-zero eigenvalues, so we get

$$\begin{aligned} r \geq \text{rank } \tilde{A}_G &= \dim [E_{-m}(\tilde{A}_G)] + \dim [E_{\lambda_1}(\tilde{A}_G)] + \dim [E_{\lambda_2}(\tilde{A}_G)] \\ &\geq (r-2) + 1 + 1 = r \end{aligned}$$

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and so $E_{-m}(\tilde{A}_G) = r-2$. Therefore, the characteristic polynomial is

$$p(x) = x^{mr-r-1}(x+m)^{r-2}(x-\lambda_1)(x-\lambda_2).$$

Proofs

And we finally get

$$\begin{aligned}\#\text{spanning trees } T_r(n) &= \det \tilde{L}_G \\ &= \det \left(m(r-1)I_{n-1} - \tilde{A}_G \right) \\ &= p(m(r-1)) \\ &= (m(r-1))^{mr-r} (mr)^{r-2}.\end{aligned}$$



Spanning Trees of $K_{m_1, \dots, m_r}$

Since $K_{m_1, \dots, m_r}$ is not regular, we cannot use the same techniques, but by finding the eigenvalues of the Laplacian we get the following result:

Proposition

The number of spanning trees of $K_{m_1, \dots, m_r}$ with $m_1 + \dots + m_r = n$ is

$$n^{r-2} \prod_{k=1}^r (n - m_k)^{m_k-1}.$$

Note this generalizes the previous result when $m_1 = \dots = m_r = m$.

Sketch of Proof

We again guess eigenvectors and show that we get a full set of eigenvalues.

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There are $r - 1$ eigenvectors with eigenvalue n which are of the following form:

$$\left[\begin{array}{c} 1 \\ \vdots \\ -m_1/m_k \\ \vdots \\ 0 \end{array} \right] \begin{array}{l} \} m_1 \\ \\ \} m_k \end{array}$$

Sketch of Proof

For each k there are $m_k - 1$ eigenvectors of eigenvalue $n - m_k$ of the following form:

$$\begin{bmatrix} 0 \\ \vdots \\ v_k \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} \} m_1 \\ \\ \} m_k, \\ \\ \} m_r \end{matrix} \quad \text{where} \quad v_k = \begin{bmatrix} 1 \\ \vdots \\ -1 \\ \vdots \\ 0 \end{bmatrix}$$

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The number of spanning trees is then:

$$\frac{1}{n} \prod_{\lambda_i \neq 0} \lambda_i = n^{r-2} \prod_{k=1}^r (n - m_k)^{m_k-1}$$

The Permutohedron

We analyze the Cayley graph for S_n when using the set of adjacent transpositions

$$T = \{(1, 2), (2, 3), \dots, (n - 1, n)\}$$

as a generating set.

The resulting graph is also the 1-skeleton of the permutohedron, an $(n - 1)$ -dimensional polytope:

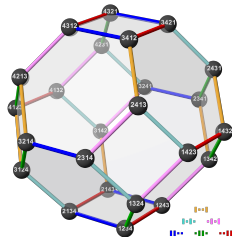


Figure: The permutohedron of order 4 (Wikipedia)

Adjacency Matrix

The recursive nature of S_n admits a nice recursive ordering of its elements:

S_1	1					
S_2	12			21		
S_3	123	132	312	213	231	321
S_4	1234 1243 1423 4123	1324 1342 1432 4132	3124 3142 3412 4312	2134 2143 2413 4213	2314 2341 2431 4231	3214 3241 3421 4321

Table: A Recursive Ordering of S_n

We will use this ordering of the vertices for the construction of the adjacency matrix of the permutohedron.

Adjacency Matrix

Using this ordering, we can construct the adjacency matrix A_n for S_n as a block matrix from A_{n-1} :

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Adjacency Matrix

Can turn this intuition into an explicit recursion using the Kronecker Product:

$$A_n = \sum_{i=1}^{n-1} B_i^{(n)}$$

Where:

$$B_0^{(n)} = B_n^{(n)} = \mathbf{0} \text{ and } B_i^{(n)} = (\text{id}_{(n-1)!} \otimes D_{n,i}) + (B_{i-1}^{(n-1)} \otimes L_{n,i}) + (B_i^{(n-1)} \otimes U_{n,i})$$

With $n \times n$ matrices:

$$D_{n,i} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad L_{n,i} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad U_{n,i} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_i \qquad \underbrace{\hspace{10em}}_{i-1} \qquad \underbrace{\hspace{10em}}_{i+1}$

Adjacency Matrix: S_3

Some pictures of the adjacency matrices and their spectra:

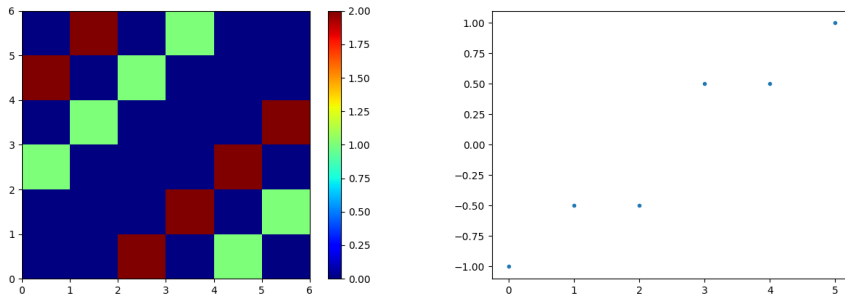


Figure: Matrix and Spectrum: color corresponds to the transposition used.

Adjacency Matrix: S_4

Some pictures of the adjacency matrices and their normalized spectra:

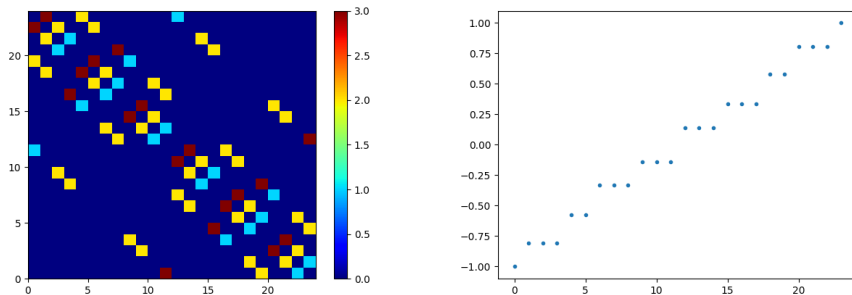


Figure: Matrix and Spectrum: color corresponds to the transposition used.

Adjacency Matrix: S_5

Some pictures of the adjacency matrices and their normalized spectra:

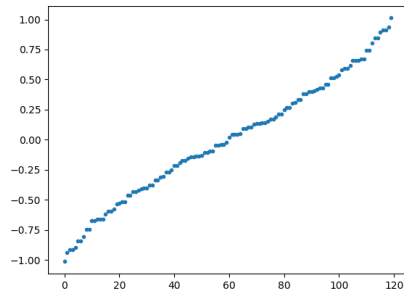
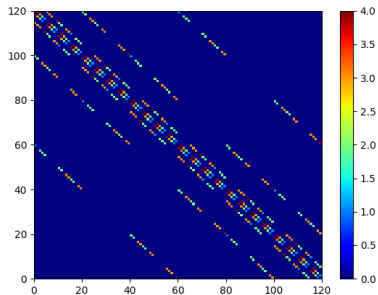


Figure: Matrix and Spectrum: color corresponds to the transposition used.

Adjacency Matrix: S_6

Some pictures of the adjacency matrices and their normalized spectra:

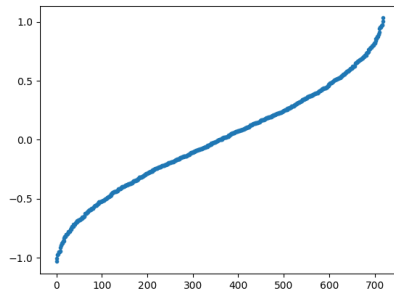
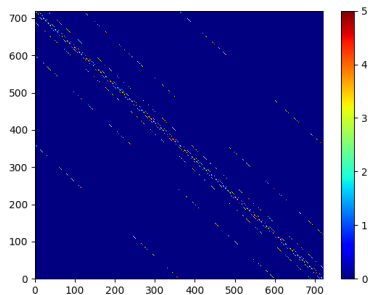


Figure: Matrix and Spectrum: color corresponds to the transposition used.

Adjacency Matrix: S_7

Some pictures of the adjacency matrices and their normalized spectra:

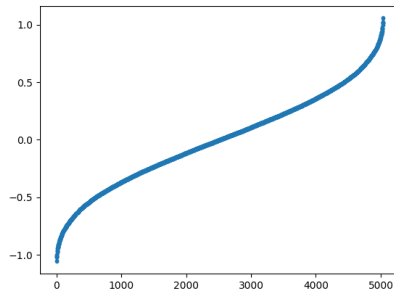
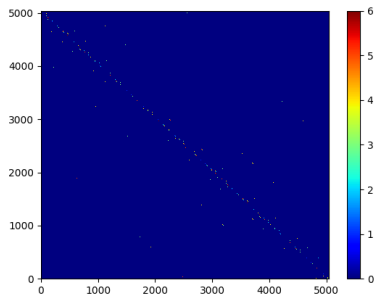


Figure: Matrix and Spectrum: color corresponds to the transposition used.

Limiting Spectrum

Not much is known about the spectrum of the permutohedron. The main explicit result is due to Bacher [1] who showed that the smallest nonzero eigenvalue of the Laplacian is $2 - 2 \cos \frac{\pi}{n}$.

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For other Cayley graphs of S_n though, much more is known. Using the generating set

$$S = \{(1, 2), (1, 3), \dots, (1, n)\},$$

the spectrum is known to be only integers [3], and the distribution of the eigenvalues approaches the semi-circle distribution [2].

Limiting Spectrum

Let A_n be the adjacency matrix of the permutohedron, and let $\lambda_1, \dots, \lambda_{n!}$ be its eigenvalues. Define

$$\mu_n = \frac{1}{n!} \sum_{i=1}^{n!} \delta_{\frac{\lambda_i}{2\sqrt{n}}}.$$

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Proposition

The measures μ_n do not converge in distribution to the semi-circle law.

Sketch of Proof

We look at the moments $m_n^{(r)}$ of μ_n . If X_n is a variable with law μ_n , then

$$\begin{aligned}
 m_n^{(r)} = \mathbb{E}[X_n^r] &= \frac{1}{n!} \sum_{i=1}^{n!} \left(\frac{\lambda_i}{2\sqrt{n}} \right)^r \\
 &= \frac{1}{2^r n^{r/2}} \left(\frac{1}{n!} \sum_{i=1}^{n!} \lambda_i^r \right) \\
 &= \frac{1}{2^r n^{r/2}} \left(\frac{1}{n!} \text{trace}(A_n^r) \right) \\
 &= \frac{(A_n^r)_{11}}{2^r n^{r/2}}
 \end{aligned}$$

Sketch of Proof

Finding $(A_n^r)_{11}$ comes down to counting how many ways to write

$$e = (i_1, i_1 + 1)(i_2, i_2 + 1) \cdots (i_r, i_r + 1).$$

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By using commutativity relations, we can show that for even r ,

$$(A_n^r)_{11} \geq n^{r/2} \left(\text{Cat} \left(\frac{r}{2} \right) + \epsilon_r \right)$$

for some $\epsilon_r > 0$.

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This implies that $m_n^{(r)}$ is larger than the moments of the semi-circle distribution, and we are done. $\square$

Future goals

Next steps Find the limiting or exact spectrum of the permutohedron.

Challenges Understand the representation theory approach to eigenvalues of Cayley graphs.

References

- [1] Roland Bacher. “Minimal Eigenvalue of the Coxeter Laplacian for the Symmetrical Group”. In: *Journal of Algebra* 167.2 (1994), pp. 460–472. ISSN: 0021-8693. DOI: <https://doi.org/10.1006/jabr.1994.1195>.
- [2] Philippe Biane. “Permutation model for semi-circular systems and quantum random walks.”. In: *Pacific Journal of Mathematics* 171.2 (1995), pp. 373–387.
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